United States Patent	12408500
Kind Code	B2
Date of Patent	September 02, 2025
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Light-emitting module and display device

Abstract

A light-emitting module and a display device are provided. Reflective units are disposed, which include a main body disposed on a backplate and a reflective part disposed on the main body. The main body is provided with a retaining wall structure on one side away from the backplate, and an orthographic projection of the reflective part on the backplate covers an orthographic projection of the main body on the backplate, thereby effectively blocking a flow of a material of the reflective part, thereby controlling a height of the reflective part.

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Appl. No.:	**17/756659**
Filed (or PCT Filed):	**May 16, 2022**
PCT No.:	**PCT/CN2022/093057**
PCT Pub. No.:	**WO2023/201803**
PCT Pub. Date:	October 26, 2023

Prior Publication Data

Document Identifier	**Publication Date**
US 20240162387 A1	May. 16, 2024

Foreign Application Priority Data

CN	202210410863.0	Apr. 19, 2022

Publication Classification

Int. Cl.: **H10H20/856** (20250101); **H01L25/075** (20060101); **H10H20/841** (20250101)

U.S. Cl.:

CPC **H10H20/856** (20250101); **H01L25/0753** (20130101); **H10H20/841** (20250101);

Field of Classification Search

USPC: None

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Background/Summary

FIELD OF INVENTION

(1) The present disclosure relates to the field of display technologies, and more particularly, to a light-emitting module and a display device.

BACKGROUND OF INVENTION

(2) In mini-light-emitting diode (mini-LED) displays, a large number of densely distribution is usually used to realize dimming in a small area. Therefore, under the premise of small light-mixing distances, they can still achieve good brightness uniformity, and pictures are colorful and have higher contrast. With high color gamut configuration, effects thereof can be comparable to organic light-emitting diodes (OLEDs). Meanwhile, an existence of a large number of dense light sources is easier to achieve higher brightness requirements than conventional LED displays. At present, in view of many advantages of mini-LED displays, the mini-LED displays have significant advantages in applications of ultra-thin, high color rendering, and power saving, and have received extensive attention from major manufacturers.

(3) Conventional mini-LED light-emitting modules usually include reflective coating layers. The reflective coating layers are provided with windows, and mini-light-emitting diodes are disposed in the windows. The reflective coating layers can effectively reflect light emitted from lateral sides of the mini-light-emitting diodes to a light-emitting surface of the light-emitting modules to improve emitting efficiency of the light. However, the reflective coating layers usually use white oil material that has fluidity. Therefore, in practical manufacturing processes, a height of the reflective coating layers cannot be effectively controlled, thereby causing problems such as difficulty in fabrication processes and inability to effectively reflect the light emitted from the lateral sides of the mini-light-emitting diodes to the light-emitting surface of the light-emitting modules.

(4) Technical problem: an embodiment of the present disclosure provides a light-emitting module and a display device to solve deficiencies in related technologies.

SUMMARY OF INVENTION

(5) In order to solve the above technical problems, the embodiments of the present disclosure provide technical solutions as follows.

(6) An embodiment of the present disclosure provides a light-emitting module, which includes: a backplate; a plurality of light-emitting units disposed on the backplate; and a reflective layer disposed on the backplate and including a plurality of reflective units, wherein, one of the reflective units surrounds one of the light-emitting units, and the reflective units include a reflective surface on one side away from the backplate; wherein, in one light-emitting unit and one reflective unit corresponding to the light-emitting unit, a height of the reflective surface is gradually increased along a direction away from the light-emitting unit; and the reflective unit includes a main body disposed on the backplate and a reflective part disposed on the main body, the reflective surface is located on one side of the reflective part away from the main body, the main body is provided with a retaining wall structure on one side away from the backplate, and an orthographic projection of the reflective part on the backplate covers an orthographic projection of the main body on the backplate.

(7) In the light-emitting module provided in the embodiment of the present disclosure, in the reflective unit and the light-emitting unit corresponding to the reflective unit, a height of the main body is gradually increased along the direction away from the light-emitting unit.

(8) In the light-emitting module provided in the embodiment of the present disclosure, in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body includes a plurality of raised parts disposed in sequence away from the light-emitting unit, and heights of the raised parts are gradually increased along the direction away from the light-emitting unit.

(9) In the light-emitting module provided in the embodiment of the present disclosure, the main body at least includes a first raised part and a second raised part disposed in sequence away from the light-emitting unit; and the second raised part includes a raised main part and the retaining wall structure disposed in a stack, a height of the raised main part is greater than a height of the first raised part, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the first raised part in the direction perpendicular to the backplate.

(10) In the light-emitting module provided in the embodiment of the present disclosure, the main body at least includes a first raised part and a second raised part disposed in sequence away from the light-emitting unit, and an orthographic projection of the first raised part on the backplate and an orthographic projection of the second raised part on the backplate do not overlap; and the retaining wall structure is located on one side of the second raised part away from the backplate, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the second raised part in the direction perpendicular to the backplate.

(11) In the light-emitting module provided in the embodiment of the present disclosure, orthographic projections of adjacent raised parts on the backplate do not overlap; the retaining wall structure includes a plurality of retaining walls, and one of the retaining walls corresponds to one of the raised parts; and in a direction perpendicular to the backplate, heights of adjacent retaining walls are equal.

(12) In the light-emitting module provided in the embodiment of the present disclosure, in the direction away from the light-emitting unit, gaps between the adjacent retaining walls are equal or are gradually increased.

(13) In the light-emitting module provided in the embodiment of the present disclosure, in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body includes a raised part, and the raised part includes a raised beveled surface on one side away from the backplate; and in the reflective unit and the light-emitting unit corresponding to the reflective unit, a height of the raised part is gradually increased along the direction away from the light-emitting unit, the retaining wall structure includes a plurality of retaining walls disposed on the raised beveled surface and surrounding the light-emitting unit, and the retaining walls are arranged in sequence along a direction from the reflective unit to the light-emitting unit.

(14) In the light-emitting module provided in the embodiment of the present disclosure, in a direction perpendicular to the backplate, in any two of the retaining walls, a distance from one of the retaining walls away from the light-emitting unit to the backplate is greater than a distance from another one of the retaining walls adjacent to the light-emitting unit to the backplate.

(15) In the light-emitting module provided in the embodiment of the present disclosure, an included angle between a tangent line of any point on the reflective surface and the backplate is greater than or equal to 60 degrees.

(16) An embodiment of the present disclosure provides a display device, which includes a display panel and a light-emitting module, wherein, the display panel includes a display area and a non-display area adjacent to the display area, and the light-emitting module is disposed corresponding to the display area and includes: a backplate; a plurality of light-emitting units disposed on the

backplate; and a reflective layer disposed on the backplate and including a plurality of reflective units, wherein, one of the reflective units surrounds one of the light-emitting units, and the reflective units include a reflective surface on one side away from the backplate; wherein, in one light-emitting unit and one reflective unit corresponding to the light-emitting unit, a height of the reflective surface is gradually increased along a direction away from the light-emitting unit; and the reflective unit includes a main body disposed on the backplate and a reflective part disposed on the main body, the reflective surface is located on one side of the reflective part away from the main body, the main body is provided with a retaining wall structure on one side away from the backplate, and an orthographic projection of the reflective part on the backplate covers an orthographic projection of the main body on the backplate.

(17) In the display device provided in the embodiment of the present disclosure, in the reflective unit and the light-emitting unit corresponding to the reflective unit, a height of the main body is gradually increased along the direction away from the light-emitting unit.

(18) In the display device provided in the embodiment of the present disclosure, in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body includes a plurality of raised parts disposed in sequence away from the light-emitting unit, and heights of the raised parts are gradually increased along the direction away from the light-emitting unit.

(19) In the display device provided in the embodiment of the present disclosure, the main body at least includes a first raised part and a second raised part disposed in sequence away from the light-emitting unit; and the second raised part includes a raised main part and the retaining wall structure disposed in a stack, a height of the raised main part is greater than a height of the first raised part, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the first raised part in the direction perpendicular to the backplate.

(20) In the display device provided in the embodiment of the present disclosure, the main body at least includes a first raised part and a second raised part disposed in sequence away from the light-emitting unit, and an orthographic projection of the first raised part on the backplate and an orthographic projection of the second raised part on the backplate do not overlap; and the retaining wall structure is located on one side of the second raised part away from the backplate, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the second raised part in the direction perpendicular to the backplate.

(21) In the display device provided in the embodiment of the present disclosure, orthographic projections of adjacent raised parts on the backplate do not overlap; the retaining wall structure includes a plurality of retaining walls, and one of the retaining walls corresponds to one of the raised parts; and in a direction perpendicular to the backplate, heights of adjacent retaining walls are equal.

(22) In the display device provided in the embodiment of the present disclosure, in the direction away from the light-emitting unit, gaps between the adjacent retaining walls are equal or are gradually increased.

(23) In the display device provided in the embodiment of the present disclosure, in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body includes a raised part, and the raised part includes a raised beveled surface on one side away from the backplate; and in the reflective unit and the light-emitting unit corresponding to the reflective unit, a height of the raised part is gradually increased along the direction away from the light-emitting unit, the retaining wall structure includes a plurality of retaining walls disposed on the raised beveled surface and surrounding the light-emitting unit, and the retaining walls are arranged in sequence along a direction from the reflective unit to the light-emitting unit.

(24) In the display device provided in the embodiment of the present disclosure, in a direction perpendicular to the backplate, in any two of the retaining walls, a distance from one of the

retaining walls away from the light-emitting unit to the backplate is greater than a distance from another one of the retaining walls adjacent to the light-emitting unit to the backplate.
(25) In the display device provided in the embodiment of the present disclosure, an included angle between a tangent line of any point on the reflective surface and the backplate is greater than or equal to 60 degrees.
(26) Beneficial effect: the embodiments of the present disclosure provide the light-emitting module and the display device. The light-emitting module includes the backplate and the plurality of light-emitting units and the reflective layer disposed on the backplate. The reflective layer includes the plurality of reflective units, one of the reflective units surrounds one of the light-emitting units, and the reflective units include a reflective surface on one side away from the backplate. In one light-emitting unit and one reflective unit corresponding to the light-emitting unit, the height of the reflective surface is gradually increased along the direction away from the light-emitting unit. The embodiments of the present disclosure dispose the reflective units, which include the main body disposed on the backplate and the reflective part disposed on the main body. One side of the main body away from the backplate includes the retaining wall structure, and the orthographic projection of the reflective part on the backplate covers the orthographic projection of the main body on the backplate, thereby effectively blocking a flow of a material of the reflective part, thereby controlling a height of the reflective part.

Description

DESCRIPTION OF DRAWINGS
(1) The following detailed description of specific embodiments of the present disclosure will make the technical solutions and other beneficial effects of the present disclosure obvious with reference to the accompanying drawings.
(2) FIG. **1** is a first schematic structural diagram of a light-emitting module in current technology.
(3) FIG. **2** is a second schematic structural diagram of the light-emitting module in current technology.
(4) FIG. **3** is a first schematic cross-sectional diagram of a light-emitting module according to an embodiment of the present disclosure.
(5) FIG. **4** is a first schematic cross-sectional top view of the light-emitting module according to an embodiment of the present disclosure.
(6) FIG. **5** is a second schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure.
(7) FIG. **6** is a third schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure.
(8) FIG. **7** is a second schematic cross-sectional top view of the light-emitting module according to an embodiment of the present disclosure.
(9) FIG. **8** is a fourth schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure.
(10) FIG. **9** is a fifth schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS
(11) The embodiments of the present disclosure provide a light-emitting module and a display device. In order to make the purpose, technical solutions, and effects of the present disclosure clearer and more definite, the following further describes the present disclosure in detail with reference to the drawings and embodiments. It should be understood that the specific embodiments described herein are only used to explain the disclosure and are not used to limit the disclosure.
(12) Referring to FIGS. **3** to **8**, an embodiment of the present disclosure provides the light-emitting

module and the display device. The light-emitting module **1** includes: a backplate **10**; a plurality of light-emitting units **20** disposed on the backplate **10**; and a reflective layer **40** disposed on the backplate **10** and including a plurality of reflective units **400**, wherein, one of the reflective units **400** surrounds one of the light-emitting units **20**, and the reflective units **400** include a reflective surface **400**A on one side away from the backplate **10**.
(13) In one light-emitting unit **20** and one reflective unit **400** corresponding to the light-emitting unit **20**, a height of the reflective surface **400**A is gradually increased along a direction away from the light-emitting unit **20**.
(14) The reflective unit **400** includes a main body **410** disposed on the backplate **10** and a reflective part **420** disposed on the main body **410**. The main body **410** is provided with a retaining wall structure **430** on one side away from the backplate **10**. The reflective surface **400**A is located on one side of the reflective part **420** away from the main body **410**, and an orthographic projection of the reflective part **420** on the backplate **10** covers an orthographic projection of the main body **410** on the backplate **10**.
(15) It can be understood that referring to FIG. **1**, FIG. **1** is a first schematic structural diagram of a light-emitting module in current technology. A conventional mini-light-emitting diode (mini-LED) light-emitting module **1** usually includes a backplate **10** and a plurality of light-emitting units **20** on the backplate **10**. The light-emitting units **20** include but are not limited to mini-light-emitting diodes. In order to improve reflectivity of the light-emitting units **20**, a reflective layer **40** is usually disposed on a surface of the backplate **10**. The reflective layer **40** is disposed surrounding peripheries of the light-emitting units **20**. A material of the reflective layer **40** includes but is not limited to white oil, and reflectivity of reflective units can be improved by the reflection effect of the white oil. However, light is reflected in all directions, and conventional white oil substrate is in a flat state, which will cause partial light laterally emitted from the light-emitting units **20** to have a larger light-emitting angle, thereby being unable to reach the observation surface (that is, having limited light-concentrating effect).
(16) In order to solve the above problem, referring to FIG. **2**, FIG. **2** is a second schematic structural diagram of the light-emitting module **1** in current technology. The light-emitting module **1** includes a backplate **10** and a plurality of light-emitting units **20** and a reflective layer **40** disposed on the backplate **10**. The reflective layer **40** is provided with windows (not marked in the figure), and the light-emitting units **20** are located in the windows. A surface of the windows facing the light-emitting units **20** is an inclined surface relative to the backplate **10**, so the reflective layer **40** can effectively reflect the light emitted laterally from the light-emitting units **20** to the light-emitting surface of the light-emitting module **1**, thereby improving the light-emitting efficiency. However, the reflective layer **40** usually uses white oil material that has fluidity. Therefore, in practical manufacturing processes, a height of the reflective layer **40** cannot be effectively controlled, thereby causing problems such as difficulty in fabrication processes and inability to effectively reflect the light emitted from lateral sides of the light-emitting units **20** to the light-emitting surface of the light-emitting module **1**.
(17) In view of foregoing problems, in the embodiments of the present disclosure, in one light-emitting unit **20** and one reflective unit **400** corresponding to the light-emitting unit **20**, a height of the reflective surface **400**A is set to gradually increase along the direction away from the light-emitting unit **20**. The reflective unit **400** includes the main body **410** disposed on the backplate **10** and the reflective part **420** disposed on the main body **410**. The main body **410** is provided with the retaining wall structure **430** on one side away from the backplate **10**. The reflective surface **400**A is located on one side of the reflective part **420** away from the main body **410**, and the orthographic projection of the reflective part **420** on the backplate **10** covers the orthographic projection of the main body **410** on the backplate **10**. Therefore, a flow of a material of the reflective part **420** can be effectively blocked and the height of the reflective part **420** can be controlled, thereby further maintaining a height of the reflective surface **400**A and allowing the light emitted laterally from the

light-emitting units **20** to be effectively reflected to the light-emitting surface of the light-emitting module **1**, thereby improving the light-emitting efficiency.

(18) Specific embodiments are used to describe technical solutions of the present disclosure.

(19) In an embodiment, referring to FIGS. **3** and **4**, FIG. **3** is a first schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure, and FIG. **4** is a first schematic cross-sectional top view of the light-emitting module according to an embodiment of the present disclosure.

(20) This embodiment provides the light-emitting module **1** for providing backlight. The light-emitting module **1** includes the backplate **10** and the plurality of light-emitting units **20** and the reflective layer **40** disposed on the backplate **10**.

(21) The backplate **10** includes but is not limited to a circuit board, and the light-emitting units **20** include but are not limited to mini-light-emitting diodes (mini-LEDs), and the backplate **10** is used to provide driving electrical signals for the light-emitting units **20**. Each of the light-emitting units **20** includes two electrodes **21** on one side adjacent to the backplate **10**, and the backplate **10** includes two bonding pads (not shown in the figures) corresponding to the two electrodes **21** on one side adjacent to the light-emitting units **20**. The light-emitting units **20** are welded on the backplate **10** by processes such as reflow soldering, and the electrodes **21** are electrically connected to the bonding pads by solder paste. It should be noted that the backplate **10** being the circuit board and the electrodes **21** being electrically connected to the bonding pads by the solder paste are only taken as an example for description. The type of the backplate **10** and the connection method between the electrodes **21** and the bonding pads are not specifically limited in the embodiments.

(22) The reflective layer **40** includes the plurality of reflective units **400**, one of the reflective units **400** surrounds one of the light-emitting units **20**, and the reflective units **400** include the reflective surface **400**A on one side away from the backplate **10**. In one light-emitting unit **20** and one reflective unit **400** corresponding to the light-emitting unit **20**, the height of the reflective surface **400**A is gradually increased along the direction away from the light-emitting unit **20**.

(23) The reflective unit **400** includes the main body **410** disposed on the backplate **10** and the reflective part **420** disposed on the main body **410**, the reflective surface **400**A is located on one side of the reflective part **420** away from the main body **410**, the main body **410** is provided with the retaining wall structure **430** on one side away from the backplate **10**, and the orthographic projection of the reflective part **420** on the backplate **10** covers the orthographic projection of the main body **410** on the backplate **10**.

(24) In the embodiments, a material of the main body **410** includes but is not limited to transparent photosensitive materials such as photo-spacer (PS) color resists, over-coat resins (OC), and array organic insulating films (polymer film on array, PFA), or other photosensitive materials such as RGB photoresist materials. The reflective part **420** has a reflective effect, which can reflect the light emitted from the light-emitting units **20** toward the backplate **10** to the light-emitting side of the light-emitting module **1**, thereby improving the light utilization efficiency. A material of the reflective part **420** includes but is not limited to materials with reflective properties, such as white oil. In this embodiment, after disposing the light-emitting units **20** on the backplate **10**, the main body **410** is disposed surrounding the light-emitting unit **20**, and then the retaining wall structure **430** is disposed on one side of the main body **410** away from the backplate **10**. A layer of white oil is coated on the main body **410** to form the reflective units **400**, and a height of the reflective units **400** is less than a height of the light-emitting units **20**. That is, in the direction perpendicular to the backplate **10**, in one light-emitting unit **20** and one reflective unit **400** corresponding to the light-emitting unit **20**, a distance from any point on the reflective layer **40** to the backplate **10** is less than the height of the light-emitting unit **20**.

(25) It can be understood that in the embodiments, one reflective unit **400** is set to surround one light-emitting unit **20**, and the reflective unit **400** includes the reflective surface **400**A on the side away from the backplate **10**. In one light-emitting unit **20** and one reflective unit **400** corresponding

to the light-emitting unit **20**, the height of the reflective surface **400**A is gradually increased along the direction away from the light-emitting unit **20**. The reflective surface **400**A is used to receive the light emitted laterally from the light-emitting unit **20** and effectively reflect the light emitted laterally from the light-emitting units **20** to the light-emitting surface of the light-emitting module **1**, thereby improving the light-emitting efficiency. In addition, each of the reflective units **400** is set to include the main body **410** disposed on the backplate **10** and the reflective part **420** disposed on the main body **410**. The reflective surface **400**A is set to be located on the side of the reflective part **420** away from the main body **410**, the main body **410** is set to be provided with the retaining wall structure **430** on the side away from the backplate **10**, and the orthographic projection of the reflective part **420** on the backplate **10** is set to cover the orthographic projection of the main body **410** on the backplate **10**. The retaining wall structure **430** can effectively block the flow of the material of the reflective part **420**, thereby controlling the height of the reflective part **420**. Therefore, in practical manufacturing processes, the height of the reflective layer **40** being unable to be effectively controlled can be prevented, thereby preventing the problems such as difficulty in fabrication processes and inability to effectively reflect the light emitted from the lateral sides of the light-emitting units **20** to the light-emitting surface of the light-emitting module **1**.

(26) In this embodiment, in the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, a height of the main body **410** is gradually increased along the direction away from the light-emitting unit **20**. Specifically, the main body **410** includes a plurality of raised parts disposed in sequence away from the light-emitting unit **20**, and heights of the raised parts are gradually increased along the direction away from the light-emitting unit **20**. Wherein, the raised parts are disposed on the backplate **10** and protrude from a body surface toward the side away from the backplate **10**.

(27) It can be understood that in this embodiment, the main body **410** is set to include the plurality of raised parts disposed in sequence away from the light-emitting unit **20**, and the heights of the raised parts are set to be gradually increased along the direction away from the light-emitting unit **20**, thereby forming the reflective surface **400**A on the side of the reflective units **400** away from the backplate **10**. Further, the light emitted laterally from the light-emitting unit **20** can be effectively reflected to the light-emitting surface of the light-emitting module **1** by the reflective surface **400**A, thereby improving the light-emitting efficiency.

(28) Preferably, the main body **410** at least includes a first raised part **411** and a second raised part **412** disposed in sequence away from the light-emitting unit **20**. The second raised part **412** includes a raised main part and the retaining wall structure **430** disposed in a stack, and an orthographic projection of the retaining wall structure **430** in a direction perpendicular to the backplate **10** is within an orthographic projection of the first raised part **411** in the direction perpendicular to the backplate **10**. Specifically, in this embodiment, the main body **410** including the first raised part **411**, the second raised part **412**, a third raised part **413**, and a fourth raised part **414** disposed in sequence away from the light-emitting unit **20** is taken as an example for description.

(29) In this embodiment, the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** are disposed adjacent to each other in sequence. Wherein, the retaining wall structure **430** includes a first retaining wall **431**, a second retaining wall **432**, and a third retaining wall **433** disposed around the light-emitting unit **20**.

(30) The second raised part **412** includes a first raised main part **412**A and the first retaining wall **431** disposed in a stack, the third raised part **413** includes a second raised main part **413**A and the second retaining wall **432** disposed in a stack, and the fourth raised part **414** includes a third raised main part **414**A and the third retaining wall **433** disposed in a stack. That is, the first raised main part **412**A and the first retaining wall **431** are integrally formed, the second raised main part **413**A and the second retaining wall **432** are integrally formed, and the third raised main part **414**A and the third retaining wall **433** are integrally formed.

(31) Wherein, a height of the third raised main part **414**A is greater than a height of the third raised

part **413**, a height of the second raised main part **413**A is greater than a height of the second raised part **412**, and a height of the first raised main part **412**A is greater than a height of the first raised part **411**.
(32) Further, an orthographic projection of the second raised part **412** on the backplate **10** at least partially overlaps an orthographic projection of the first raised part **411** on the backplate **10**; an orthographic projection of the third raised part **413** on the backplate **10** at least partially overlaps an orthographic projection of the second raised part **412** on the backplate **10**; and an orthographic projection of the fourth raised part **414** on the backplate **10** at least partially overlaps an orthographic projection of the third raised part **413** on the backplate **10**. Specifically, an orthographic projection of the first retaining wall **431** in the direction perpendicular to the backplate **10** is within an orthographic projection of the first raised part **411** in the direction perpendicular to the backplate **10**; an orthographic projection of the second retaining wall **432** in the direction perpendicular to the backplate **10** is within an orthographic projection of the second raised part **412** in the direction perpendicular to the backplate **10**; and an orthographic projection of the third retaining wall **433** in the direction perpendicular to the backplate **10** is within an orthographic projection of the third raised part **413** in the direction perpendicular to the backplate **10**.
(33) It should be noted that from common knowledge, it can be known that in color filters, when two adjacent color resists with different colors overlap, a raised portion that looks like a horn will form. Therefore, in this embodiment, using the above principle, the first raised main part **412**A is set to overlap the first raised part **411**, and the height of the first raised main part **412**A is greater than the height of the first raised part **411**, thereby forming the first retaining wall **431** at a position where the first raised main part **412**A overlaps the first raised part **411**. The second raised main part **413**A is set to overlap the second raised part **412**, and the height of the second raised main part **413**A is greater than the height of the second raised part **412**, thereby forming the second retaining wall **432** at a position where the second raised main part **413**A overlaps the second raised part **412**. The third raised main part **414**A is set to overlap the third raised part **413**, and the height of the third raised main part **414**A is greater than the height of the third raised part **413**, thereby forming the third retaining wall **433** at a position where the third raised main part **414**A overlaps the third raised part **413**.
(34) Therefore, in this embodiment, by disposing the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** in sequence and making two adjacent raised parts overlap with each other, retaining walls that look like horns can be formed, thereby further saving manufacturing process steps of the light-emitting module **1**. Meanwhile, the retaining walls that look like horns can effectively block the flow of the material of the reflective part **420** located on the main body **410**, thereby controlling the height of the reflective part **420**.
(35) In this embodiment, the height and width of the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** are not specifically limited.
(36) Further, in this embodiment, the orthographic projection of the reflective part **420** on the backplate **10** covers the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414**, and an included angle between a tangent line of any point on the reflective surface **400**A and the backplate **10** is greater than or equal to 60 degrees. Specifically, the reflective surface **400**A includes a first portion (not marked in the figures), and in the direction perpendicular to the backplate **10**, a cross-section of the first portion is a straight line or an arc. Preferably, in the direction perpendicular to the backplate **10**, the cross-section of the first portion is straight, and the first portion is an inclined plane.
(37) Further, the reflective surface **400**A further includes other portions, and in the direction perpendicular to the backplate **10**, a cross-section of the other portions may be a straight line or an arc, which is not specifically limited in this embodiment.
(38) It can be understood that an included angle between a tangent line of any point on the first

portion and the backplate is greater than or equal to 60 degrees. A size of the included angle α may be controlled by controlling the height of the main body **410**, so that the reflective part **420** may have different types of reflective surfaces **400**A. In the one reflective unit **400** and the one light-emitting unit **20** corresponding to the reflective unit **400**, the reflective surface **400**A may cover all or part of a facade of the light-emitting unit **20**, thereby further adjusting overall luminous efficiency of the light-emitting module **1**.
(39) It should be noted that referring to FIG. **5**, FIG. **5** is a second schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure. In another embodiment, the reflective surface **400**A includes the first portion (not marked in the figures), and in the direction perpendicular to the backplate **10**, the cross-section of the first portion is a straight line or an arc. Preferably, in the direction perpendicular to the backplate **10**, the cross-section of the first portion is arc-shaped, and the first portion is a concave curved surface.
(40) Further, in another embodiment, the included angle between the tangent line of any point on the reflective surface **400**A and the backplate is greater than or equal to 60 degrees. Specifically, in the direction perpendicular to the backplate **10**, the cross-section of the reflective surface **400**A is a straight line or an arc, which is not specifically limited in this embodiment.
(41) In summary, in this embodiment, by disposing the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** in sequence and making two adjacent raised parts overlap with each other, the retaining walls that look like horns can be formed, thereby further saving the manufacturing process steps of the light-emitting module **1**. Meanwhile, the retaining walls that look like horns can effectively block the flow of the material of the reflective part **420** located on the main body **410**, thereby controlling the height of the reflective part **420**. In addition, the reflective part **420** may have different types of reflective surfaces **400**A by controlling the height of the reflective part **420**. In the one reflective unit **400** and the one light-emitting unit **20** corresponding to the reflective unit **400**, the reflective surface **400**A may cover all or part of the facade of the light-emitting unit **20**, thereby further improving the overall luminous efficiency of the light-emitting module **1**.
(42) It should be noted that in this embodiment, the reflective units **400** may be disposed at intervals or adjacent to each other, which is not specifically limited herein.
(43) Referring to FIGS. **6** and **7**, FIG. **6** is a third schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure, and FIG. **7** is a second schematic cross-sectional top view of the light-emitting module according to an embodiment of the present disclosure.
(44) In this embodiment, a structure of the light-emitting module **1** is basically similar to or same as that of the light-emitting module **1** provided in the above embodiment, and for details, please refer to the description of the light-emitting module **1** in the above embodiment, which is not repeated herein. An only difference between the two is:
(45) in this embodiment, the main body **410** at least includes the first raised part **411** and the second raised part **412** disposed in sequence away from the light-emitting unit **20**, and the orthographic projection of the first raised part **411** on the backplate **10** and the orthographic projection of the second raised part **412** on the backplate **10** do not overlap; and the retaining wall structure **430** is located on one side of the second raised part **412** away from the backplate **10**, and the orthographic projection of the retaining wall structure **430** in a direction perpendicular to the backplate **10** is within an orthographic projection of the second raised part **412** in the direction perpendicular to the backplate **10**.
(46) Specifically, in this embodiment, the main body **410** including the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** disposed in sequence away from the light-emitting unit **20** is taken as an example for description.
(47) In this embodiment, the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** are disposed adjacent to each other in sequence. a height of the

fourth raised part **414** is greater than a height of the third raised part **413**, a height of the third raised part **413** is greater than a height of the second raised part **412**, and a height of the second raised part **412** is greater than a height of the first raised part **411**. Wherein, the retaining wall structure **430** includes the first retaining wall **431**, the second retaining wall **432**, and the third retaining wall **433** disposed around the light-emitting unit **20**. It should be noted that a material of the retaining wall structure **430** may be same as or different from the material of the main body **410**, which is not specifically limited in this embodiment.
(48) Preferably, the first retaining wall **431** is located on one side of the second raised part **412** away from the backplate **10**, the second retaining wall **432** is located on one side of the third raised part **413** away from the backplate **10**, and the third retaining wall **433** is located on one side of the fourth raised part **414** away from the backplate **10**. Specifically, the orthographic projection of the first retaining wall **431** in the direction perpendicular to the backplate **10** is within the orthographic projection of the second raised part **412** in the direction perpendicular to the backplate **10**; the orthographic projection of the second retaining wall **432** in the direction perpendicular to the backplate **10** is within the orthographic projection of the third raised part **413** in the direction perpendicular to the backplate **10**; and the orthographic projection of the third retaining wall **433** in the direction perpendicular to the backplate **10** is within an orthographic projection of the fourth raised part **414** in the direction perpendicular to the backplate **10**. The positions of the first retaining wall **431**, the second retaining wall **432**, and the third retaining wall **433** are not specifically limited in this embodiment.
(49) It can be understood that in this embodiment, the main body **410** is set to include the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** disposed in sequence away from the light-emitting unit **20**. The retaining wall structure **430** includes the first retaining wall **431**, the second retaining wall **432**, and the third retaining wall **433** disposed around the light-emitting unit **20**. The first retaining wall **431** is located on the second raised part **412**, the second retaining wall **432** is located on the third raised part **413**, and the third retaining wall **433** is located on the fourth raised part **414**. Therefore, the flow of the material of the reflective part **420** located on the main body **410** can be effectively blocked, thereby controlling the height of the reflective part **420**.
(50) In this embodiment, the retaining wall structure **430** further includes a fourth retaining wall **434** disposed around the light-emitting units **20**, the fourth retaining wall **434** is located on one side of the first raised part **411** away from the backplate **10**, and an orthographic projection of the fourth retaining wall **434** in the direction perpendicular to the backplate **10** is within the orthographic projection of the first raised part **411** in the direction perpendicular to the backplate **10**. Compared to the above embodiments, in this embodiment, the first retaining wall **431**, the second retaining wall **432**, the third retaining wall **433**, and the fourth retaining wall **434** are disposed, which can further stop the flow of the material of the reflective part **420** located on the main body **410**, thereby controlling the height of the reflective part **420**. It can be understood that a number of the retaining walls is not specifically limited in this embodiment.
(51) It should be noted that widths of the first retaining wall **431**, the second retaining wall **432**, the third retaining wall **433**, and the fourth retaining wall **434** are not specifically limited in this embodiment.
(52) Referring to FIG. **8**, FIG. **8** is a fourth schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure.
(53) In this embodiment, a structure of the light-emitting module **1** is basically similar to or same as that of the light-emitting module **1** provided in the third schematic cross-sectional diagram, and for details, please refer to the description of the light-emitting module **1** in the above embodiment, which is not repeated herein. An only difference between the two is:
(54) in this embodiment, the main body **410** includes a plurality of raised parts disposed in sequence away from the light-emitting unit **20**, and heights of the raised parts are gradually

increased along the direction away from the light-emitting unit **20**. Orthographic projections of adjacent raised parts on the backplate **10** do not overlap. The retaining wall structure **430** includes a plurality of retaining walls **4300**, and one of the retaining walls **4300** corresponds to one of the raised parts. Wherein, in the direction perpendicular to the backplate **10**, heights of adjacent retaining walls are equal.
(55) Specifically, in this embodiment, the main body **410** including the first raised part **411**, the second raised part **412**, the third raised part **413**, and the fourth raised part **414** disposed in sequence away from the light-emitting unit **20** is taken as an example for description.
(56) In this embodiment, in the direction away from the light-emitting unit **20**, gaps between the adjacent retaining walls **4300** are equal or are gradually increased. Preferably, in this embodiment, the gaps between the adjacent retaining walls **4300** are gradually increased, as shown in FIG. **8**, H.sub.1>H.sub.2>H.sub.3, thereby allowing a slope of the main body **410** to be gentler in the direction away from the light-emitting units **20**. Therefore, it can further ensure that the reflective part **420** is gently coated on the main body **410**, which prevents the reflective layer **40** from being fractured, thereby being beneficial to improve the product yield.
(57) Referring to FIG. **9**, FIG. **9** is a fifth schematic cross-sectional diagram of the light-emitting module according to an embodiment of the present disclosure.
(58) In this embodiment, a structure of the light-emitting module **1** is basically similar to or same as that of the light-emitting module **1** provided in the third schematic cross-sectional diagram, and for details, please refer to the description of the light-emitting module **1** in the above embodiment, which is not repeated herein. An only difference between the two is:
(59) in this embodiment, in the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, the main body **410** includes a raised part, and the raised part includes a raised beveled surface **415**A on one side away from the backplate **10**. Wherein, in the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, a height of the raised part is gradually increased along the direction away from the light-emitting unit **20**, the retaining wall structure **430** includes a plurality of retaining walls disposed on the raised beveled surface **415**A and surrounding the light-emitting unit **20**, and the retaining walls are arranged in sequence along a direction from the reflective unit **400** to the light-emitting unit **20**.
(60) Specifically, in this embodiment, in the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, the main body **410** includes a fifth raised part **415**, and the fifth raised part **415** includes a raised beveled surface **415**A on one side away from the backplate **10**. In the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, a height of the fifth raised part **415** is gradually increased along the direction away from the light-emitting unit **20**, the retaining wall structure **430** includes the plurality of retaining walls disposed on the raised beveled surface **415**A and surrounding the light-emitting unit **20**, and the retaining walls are arranged in sequence along the direction from the reflective unit **400** to the light-emitting unit **20**, which are taken as an example for description.
(61) In this embodiment, in the direction perpendicular to the backplate **10**, in any two of the retaining walls, a distance from one of the retaining walls away from the light-emitting unit **20** to the backplate **10** is greater than a distance from another one of the retaining walls adjacent to the light-emitting unit **20** to the backplate **10**. Heights of the retaining walls are not specifically limited in this embodiment.
(62) In this embodiment, one reflective unit **400** is disposed to surround one light-emitting unit **20**. In the light-emitting unit **20** and the reflective unit **400** corresponding to the light-emitting unit **20**, the main body **410** includes the fifth raised part **415**, the height of the fifth raised part **415** is gradually increased along the direction away from the light-emitting unit **20**, and the fifth raised part **415** includes the raised beveled surface **415**A on one side away from the backplate **10**, thereby allowing the reflective unit **400** to have the reflective surface **400**A on one side away from the backplate **10**. Wherein, the retaining wall structure **430** includes the plurality of retaining walls

disposed on the raised beveled surface **415**A and surrounding the light-emitting unit **20**, and the retaining walls are arranged in sequence along the direction from the reflective unit **400** to the light-emitting unit **20**. The retaining walls can effectively block the flow of the material of the reflective part **420**, thereby controlling the height of the reflective part **420**. Therefore, in practical manufacturing processes, the height of the reflective layer **40** being unable to be effectively controlled can be prevented, thereby preventing the problems such as difficulty in fabrication processes and inability to effectively reflect the light emitted from the lateral sides of the light-emitting units **20** to the light-emitting surface of the light-emitting module **1**.
(63) It can be understood that compared to the above embodiments, in this embodiment, in the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, the main body **410** includes the fifth raised part **415**, the height of the fifth raised part **415** is gradually increased along the direction away from the light-emitting unit **20**, and the fifth raised part **415** includes the raised beveled surface **415**A on one side away from the backplate **10**, thereby allowing the reflective unit **400** to have the reflective surface **400**A on one side away from the backplate **10**. By this setting, this embodiment can save the manufacturing process steps of the light-emitting module **1**.
(64) It should be noted that in this embodiment, in the reflective unit **400** and the light-emitting unit **20** corresponding to the reflective unit **400**, the fifth raised part **415** may be manufactured on the backplate **10**, the fifth raised part **415** is provided with a groove penetrating the main body **410** along the direction perpendicular to the backplate **10**, and the light-emitting unit **20** is disposed in the groove. The fifth raised part **415** includes the raised beveled surface **415**A on the side away from the backplate **10**, and the raised beveled surface **415**A is adjacent to the light-emitting unit **20**. Wherein, the groove includes a first opening on a side close to the backplate **10** and a second opening on a side away from the backplate **10**. A maximum width of the first opening is smaller than a maximum width of the second opening. Then, the retaining walls around the light-emitting unit **20** are prepared on the raised beveled surface **415**A, and a layer of white oil is coated onto one side of the fifth raised part **415** away from the backplate **10** to form the reflective unit **400**. The height of the reflective unit **400** is less than the height of the light-emitting unit **20**.
(65) An embodiment of the present disclosure provides a display device, which includes a display panel and the light-emitting module mentioned in the above embodiments. The display panel includes a display area and a non-display area adjacent to the display area, and the light-emitting module is disposed corresponding to the display area. The reflective layer is disposed on one side of the backplate adjacent to the display panel.
(66) In this embodiment, the display panel is disposed on the light-emitting side of the light-emitting module for displaying images, and the light-emitting module is used for providing backlight to the display panel.
(67) It should be noted that in this embodiment, the display panel includes but is not limited to liquid crystal displays (LCDs), and the light-emitting module may adopt a mini-LED light board as a light source. A size of mini-LEDs is smaller than that of traditional LEDs, and using a huge amount of mini-LEDs as a backlight source can realize finer dynamic control and improve a dynamic contrast ratio of the liquid crystal displays.
(68) It can be understood that the light-emitting module has been described in detail in the above embodiments and is not repeated herein.
(69) In specific applications, the display device may be a display screen of a smart phone, a tablet computer, a notebook computer, a smart bracelet, a smart watch, smart glasses, a smart helmet, a desktop computer, a smart TV, or a digital camera, or may even be applied to electronic devices with flexible display screens.
(70) In summary, the present disclosure provides the light-emitting module and the display device. The light-emitting module includes the backplate and the plurality of light-emitting units and the reflective layer disposed on the backplate. The reflective layer includes the plurality of reflective

units, one of the reflective units surrounds one of the light-emitting units, and the reflective units include the reflective surface on one side away from the backplate. In one light-emitting unit and one reflective unit corresponding to the light-emitting unit, the height of the reflective surface is gradually increased along the direction away from the light-emitting unit. The embodiments of the present disclosure dispose the reflective units, which include the main body disposed on the backplate and the reflective part disposed on the main body. One side of the main body away from the backplate includes the retaining wall structure, and the orthographic projection of the reflective part on the backplate covers the orthographic projection of the main body on the backplate, thereby effectively blocking the flow of the material of the reflective part, thereby controlling the height of the reflective part.
(71) The present disclosure has been described with preferred embodiments thereof. The preferred embodiments are not intended to limit the present disclosure, and it is understood that many changes and modifications to the described embodiments can be carried out without departing from the scope and the spirit of the disclosure that is intended to be limited only by the appended claims.

Claims

1. A light-emitting module, comprising: a backplate; a plurality of light-emitting units disposed on the backplate; and a reflective layer disposed on the backplate and comprising a plurality of reflective units, wherein one of the reflective units surrounds one of the light-emitting units, and the reflective units comprise a reflective surface on one side away from the backplate; wherein in one light-emitting unit and one reflective unit corresponding to the light-emitting unit, a height of the reflective surface is gradually increased along a direction away from the light-emitting unit; and the reflective unit comprises a main body disposed on the backplate and a reflective part disposed on the main body, the reflective surface is located on one side of the reflective part away from the main body, the main body is provided with a retaining wall structure on one side away from the backplate, and an orthographic projection of the reflective part on the backplate covers an orthographic projection of the main body on the backplate; wherein in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body comprises a plurality of raised parts disposed in sequence away from the light-emitting unit, and heights of the raised parts are gradually increased along the direction away from the light-emitting unit.
2. The light-emitting module according to claim 1, wherein the main body at least comprises a first raised part and a second raised part disposed in sequence away from the light-emitting unit; and the second raised part comprises a raised main part and the retaining wall structure disposed in a stack, a height of the raised main part is greater than a height of the first raised part, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the first raised part in the direction perpendicular to the backplate.
3. The light-emitting module according to claim 1, wherein the main body at least comprises a first raised part and a second raised part disposed in sequence away from the light-emitting unit, and an orthographic projection of the first raised part on the backplate and an orthographic projection of the second raised part on the backplate do not overlap; and the retaining wall structure is located on one side of the second raised part away from the backplate, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the second raised part in the direction perpendicular to the backplate.
4. The light-emitting module according to claim 1, wherein orthographic projections of adjacent raised parts on the backplate do not overlap; the retaining wall structure comprises a plurality of retaining walls, and one of the retaining walls corresponds to one of the raised parts; and in a direction perpendicular to the backplate, heights of adjacent retaining walls are equal.
5. The light-emitting module according to claim 4, wherein in the direction away from the light-emitting unit, gaps between the adjacent retaining walls are equal or are gradually increased.

6. The light-emitting module according to claim 1, wherein an included angle between a tangent line of any point on the reflective surface and the backplate is greater than or equal to 60 degrees.
7. A display device, comprising a display panel and a light-emitting module, wherein the display panel comprises a display area and a non-display area adjacent to the display area, and the light-emitting module is disposed corresponding to the display area and comprises: a backplate; a plurality of light-emitting units disposed on the backplate; and a reflective layer disposed on the backplate and comprising a plurality of reflective units, wherein one of the reflective units surrounds one of the light-emitting units, and the reflective units comprise a reflective surface on one side away from the backplate; wherein in one light-emitting unit and one reflective unit corresponding to the light-emitting unit, a height of the reflective surface is gradually increased along a direction away from the light-emitting unit; and the reflective unit comprises a main body disposed on the backplate and a reflective part disposed on the main body, the reflective surface is located on one side of the reflective part away from the main body, the main body is provided with a retaining wall structure on one side away from the backplate, and an orthographic projection of the reflective part on the backplate covers an orthographic projection of the main body on the backplate; wherein in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body comprises a plurality of raised parts disposed in sequence away from the light-emitting unit, and heights of the raised parts are gradually increased along the direction away from the light-emitting unit.
8. The display device according to claim 7, wherein the main body at least comprises a first raised part and a second raised part disposed in sequence away from the light-emitting unit; and the second raised part comprises a raised main part and the retaining wall structure disposed in a stack, a height of the raised main part is greater than a height of the first raised part, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the first raised part in the direction perpendicular to the backplate.
9. The display device according to claim 7, wherein the main body at least comprises a first raised part and a second raised part disposed in sequence away from the light-emitting unit, and an orthographic projection of the first raised part on the backplate and an orthographic projection of the second raised part on the backplate do not overlap; and the retaining wall structure is located on one side of the second raised part away from the backplate, and an orthographic projection of the retaining wall structure in a direction perpendicular to the backplate is within an orthographic projection of the second raised part in the direction perpendicular to the backplate.
10. The display device according to claim 7, wherein orthographic projections of adjacent raised parts on the backplate do not overlap; the retaining wall structure comprises a plurality of retaining walls, and one of the retaining walls corresponds to one of the raised parts; and in a direction perpendicular to the backplate, heights of adjacent retaining walls are equal.
11. The display device according to claim 10, wherein in the direction away from the light-emitting unit, gaps between the adjacent retaining walls are equal or are gradually increased.
12. The display device according to claim 7, wherein in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body comprises a raised part, and the raised part comprises a raised beveled surface on one side away from the backplate; and in the reflective unit and the light-emitting unit corresponding to the reflective unit, a height of the raised part is gradually increased along the direction away from the light-emitting unit, the retaining wall structure comprises a plurality of retaining walls disposed on the raised beveled surface and surrounding the light-emitting unit, and the retaining walls are arranged in sequence along a direction from the reflective unit to the light-emitting unit.
13. The display device according to claim 12, wherein in a direction perpendicular to the backplate, in any two of the retaining walls, a distance from one of the retaining walls away from the light-emitting unit to the backplate is greater than a distance from another one of the retaining walls adjacent to the light-emitting unit to the backplate.

14. The display device according to claim 7, wherein an included angle between a tangent line of any point on the reflective surface and the backplate is greater than or equal to 60 degrees.

15. A light-emitting module, comprising: a backplate; a plurality of light-emitting units disposed on the backplate; and a reflective layer disposed on the backplate and comprising a plurality of reflective units, wherein one of the reflective units surrounds one of the light-emitting units, and the reflective units comprise a reflective surface on one side away from the backplate; wherein in one light-emitting unit and one reflective unit corresponding to the light-emitting unit, a height of the reflective surface is gradually increased along a direction away from the light-emitting unit; and the reflective unit comprises a main body disposed on the backplate and a reflective part disposed on the main body, the reflective surface is located on one side of the reflective part away from the main body, the main body is provided with a retaining wall structure on one side away from the backplate, and an orthographic projection of the reflective part on the backplate covers an orthographic projection of the main body on the backplate; wherein in the reflective unit and the light-emitting unit corresponding to the reflective unit, the main body comprises a raised part, and the raised part comprises a raised beveled surface on one side away from the backplate; and in the reflective unit and the light-emitting unit corresponding to the reflective unit, a height of the raised part is gradually increased along the direction away from the light-emitting unit, the retaining wall structure comprises a plurality of retaining walls disposed on the raised beveled surface and surrounding the light-emitting unit, and the retaining walls are arranged in sequence along a direction from the reflective unit to the light-emitting unit.

16. The light-emitting module according to claim 15, wherein in a direction perpendicular to the backplate, in any two of the retaining walls, a distance from one of the retaining walls away from the light-emitting unit to the backplate is greater than a distance from another one of the retaining walls adjacent to the light-emitting unit to the backplate.